Complete Course Guide

Networking Setup | Cloud Computing
Ethical Hacking | Penetration Testing

24 Courses - All Modules - Tools - Questions and Answers

For education and authorised, legal use only.

Important Notice - Ethics and Legal Use

Ethical hacking and penetration testing skills must only be used on systems you own or have **written permission** to test. Unauthorised access is illegal in most countries (for example the IT Act 2000 in India, the CFAA in the USA and the Computer Misuse Act in the UK). Practise only in your own lab or legal training platforms such as TryHackMe, Hack The Box, DVWA, OWASP Juice Shop and vendor sandboxes.

How to Use This Guide

- Courses are grouped into 4 parts, 6 courses each (24 total).
- Every course lists level, duration, prerequisites, full module list, tools and 4 Q&A.
- A recommended learning path, certification roadmap and general FAQ are at the end.

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PART 1 - NETWORKING AND NETWORK SETUP

Course 1: Networking Fundamentals

Level	Beginner	**Duration**	4 weeks
Prerequisite	None	**Modules**	9

Course Contents (All Modules)

- **Module 1:** Introduction to networks: LAN, WAN, MAN, PAN, internet
- **Module 2:** OSI and TCP/IP models, layer functions
- **Module 3:** Network devices: hubs, switches, routers, modems, access points
- **Module 4:** IPv4 addressing, subnet masks, CIDR, subnetting
- **Module 5:** IPv6 basics and transition methods
- **Module 6:** Core protocols: DHCP, DNS, ARP, ICMP, HTTP/S, FTP, SMTP
- **Module 7:** Ethernet, cabling (UTP, fibre), connectors and standards
- **Module 8:** Basic troubleshooting: ping, traceroute, ipconfig/ip, nslookup
- **Module 9:** Capstone: design and document a small network

Tools and Platforms

Packet Tracer, Wireshark, GNS3

Questions and Answers

Q1. What are the 7 layers of the OSI model?
A: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q2. What is the difference between a switch and a router?
A: A switch forwards frames inside a LAN using MAC addresses; a router forwards packets between networks using IP addresses.

Q3. How many usable hosts are in a /24 network?
A: 254 (256 addresses minus network and broadcast addresses).

Q4. What does DHCP do?
A: It automatically assigns IP address, subnet mask, gateway and DNS settings to clients.

Course 2: CCNA - Routing and Switching

Level	Beginner-Intermediate	**Duration**	10 weeks
Prerequisite	Networking Fundamentals	**Modules**	12

Course Contents (All Modules)

- **Module 1:** Network fundamentals and Cisco IOS CLI
- **Module 2:** VLANs, trunking (802.1Q), inter-VLAN routing
- **Module 3:** Spanning Tree Protocol (STP, RSTP) and EtherChannel
- **Module 4:** Static routing and default routes
- **Module 5:** OSPF single-area and multi-area
- **Module 6:** EIGRP and route redistribution basics
- **Module 7:** First-hop redundancy: HSRP, VRRP

- **Module 8:** NAT, PAT and ACLs
- **Module 9:** IP services: NTP, SNMP, Syslog, DHCP relay
- **Module 10:** Layer 2 security: port security, DHCP snooping, DAI
- **Module 11:** WAN concepts, VPN, QoS basics
- **Module 12:** Network automation overview

Tools and Platforms

Cisco Packet Tracer, GNS3, CML

Questions and Answers

Q1. What is a VLAN?
A: A logical segmentation of a switch into separate broadcast domains.

Q2. Why is STP needed?
A: It prevents Layer 2 loops and broadcast storms in redundant switched topologies.

Q3. What is the administrative distance of OSPF?
A: 110.

Q4. What does NAT overload (PAT) do?
A: Maps many private addresses to one public IP using different port numbers.

Course 3: Network Setup - Home, SOHO and Small Office

Level	Beginner	**Duration**	3 weeks
Prerequisite	Networking Fundamentals	**Modules**	11

Course Contents (All Modules)

- **Module 1:** Planning: requirements, topology, IP plan, budget
- **Module 2:** Choosing ISP link, modem, router, switch and access points
- **Module 3:** Structured cabling and rack/patch panel setup
- **Module 4:** Router initial configuration, WAN/LAN, DHCP and DNS
- **Module 5:** Wi-Fi setup: SSID, WPA3, channels, guest network
- **Module 6:** Port forwarding, static IPs and DDNS
- **Module 7:** Firewall, parental control and content filtering
- **Module 8:** VPN setup (site-to-site and remote access)
- **Module 9:** NAS, printers, CCTV and IoT onboarding
- **Module 10:** Monitoring, backups and documentation
- **Module 11:** Troubleshooting slow or dropped connections

Tools and Platforms

pfSense/OPNsense, MikroTik, OpenWrt, Wi-Fi analyser

Questions and Answers

Q1. Why create a guest Wi-Fi network?
A: It isolates visitors and IoT devices from private devices and shared data.

Q2. Which Wi-Fi security should you use today?
A: WPA3 (or WPA2-AES if WPA3 is unsupported); never WEP or open networks.

Q3. What is the benefit of a static IP for a printer or NAS?
A: The address never changes, so mapping, port rules and reservations stay valid.

Q4. What is port forwarding?
A: A router rule that sends inbound traffic on a chosen port to a specific internal host.

Course 4: Enterprise Network Design and Administration (CCNP level)

Level	Intermediate-Advanced	**Duration**	12 weeks
Prerequisite	CCNA	**Modules**	12

Course Contents (All Modules)

- **Module 1:** Enterprise architecture: hierarchical and spine-leaf design
- **Module 2:** Advanced OSPF, EIGRP and BGP
- **Module 3:** Route filtering, policy-based routing, redistribution
- **Module 4:** Multicast: IGMP, PIM
- **Module 5:** MPLS and SD-WAN concepts
- **Module 6:** Wireless enterprise: controllers, roaming, RF design
- **Module 7:** Network virtualization: VRF, VXLAN, overlay/underlay
- **Module 8:** QoS design and traffic engineering
- **Module 9:** High availability and disaster recovery
- **Module 10:** Network assurance: NetFlow, telemetry, monitoring
- **Module 11:** Infrastructure security: AAA, 802.1X, TACACS+/RADIUS
- **Module 12:** Capstone enterprise design

Tools and Platforms

Cisco CML, EVE-NG, SolarWinds, Zabbix

Questions and Answers

Q1. What is BGP used for?
A: Exchanging routing information between autonomous systems; it is the routing protocol of the internet.

Q2. What is the benefit of spine-leaf topology?
A: Predictable low latency and high east-west bandwidth with easy scaling.

Q3. What does 802.1X provide?
A: Port-based network access control that authenticates devices/users before granting access.

Q4. What is VRF?
A: Virtual Routing and Forwarding - multiple isolated routing tables on one router.

Course 5: Wireless Networking and Wi-Fi Setup

Level	Beginner-Intermediate	**Duration**	4 weeks
Prerequisite	Networking Fundamentals	**Modules**	9

Course Contents (All Modules)

- **Module 1:** Radio fundamentals: frequency, bands, channels, signal strength
- **Module 2:** 802.11 standards: a/b/g/n/ac/ax (Wi-Fi 6/6E) and Wi-Fi 7
- **Module 3:** Antenna types and coverage planning
- **Module 4:** Site survey and heat-map design

- **Module 5:** Controller-based vs cloud-managed vs autonomous APs
- **Module 6:** Wireless security: WPA2/WPA3, 802.1X, RADIUS
- **Module 7:** Roaming, band steering and QoS
- **Module 8:** Interference sources and troubleshooting
- **Module 9:** Point-to-point and mesh links

Tools and Platforms

Ekahau/NetSpot, Wi-Fi Analyzer, Unifi, Aruba

Questions and Answers

Q1. Which band gives longer range: 2.4 GHz or 5 GHz?
A: 2.4 GHz travels further; 5 GHz is faster but shorter range.

Q2. What is a site survey?
A: Measuring RF coverage and interference to place access points properly.

Q3. What is Wi-Fi 6?
A: The 802.11ax standard with better efficiency in dense environments (OFDMA, MU-MIMO).

Q4. Why use 802.1X in enterprise Wi-Fi?
A: Each user authenticates individually with credentials/certificates instead of a shared password.

Course 6: Network Automation and Programmability

Level	Intermediate	**Duration**	6 weeks
Prerequisite	CCNA + basic Python	**Modules**	9

Course Contents (All Modules)

- **Module 1:** Python for network engineers
- **Module 2:** Working with JSON, YAML and XML
- **Module 3:** REST APIs, Postman and API authentication
- **Module 4:** Netmiko, NAPALM and Paramiko
- **Module 5:** Ansible for network devices
- **Module 6:** NETCONF, RESTCONF and YANG models
- **Module 7:** Git and version control for configs
- **Module 8:** CI/CD for networks, Infrastructure as Code
- **Module 9:** Intent-based networking and telemetry

Tools and Platforms

Python, Ansible, Git, Postman, Cisco DevNet sandbox

Questions and Answers

Q1. Why automate network configuration?
A: It reduces human error, enforces consistency and saves time at scale.

Q2. What is YANG?
A: A data-modelling language that defines configuration and state data for NETCONF/RESTCONF.

Q3. What does Ansible use to describe tasks?
A: YAML playbooks.

Q4. What is idempotency?
A: Running the same automation repeatedly gives the same end state without unwanted changes.

PART 2 - CLOUD COMPUTING

Course 7: Cloud Computing Fundamentals

Level	Beginner	**Duration**	3 weeks
Prerequisite	Basic IT knowledge	**Modules**	9

Course Contents (All Modules)

- **Module 1:** What is cloud computing: NIST characteristics
- **Module 2:** Service models: IaaS, PaaS, SaaS, FaaS
- **Module 3:** Deployment models: public, private, hybrid, multi-cloud
- **Module 4:** Virtualization and hypervisors
- **Module 5:** Cloud economics, TCO and pricing models
- **Module 6:** Shared responsibility model
- **Module 7:** Cloud migration strategies (the 6 Rs)
- **Module 8:** Compliance, governance and SLAs
- **Module 9:** Overview of AWS, Azure and GCP

Tools and Platforms

AWS/Azure/GCP free tier

Questions and Answers

Q1. What are IaaS, PaaS and SaaS?
A: IaaS provides raw infrastructure, PaaS a managed platform to deploy code, SaaS ready-to-use software.

Q2. What is the shared responsibility model?
A: The provider secures the cloud itself; the customer secures what they put in it (data, identities, configs).

Q3. What is elasticity?
A: Automatically scaling resources up or down with demand.

Q4. What is hybrid cloud?
A: A mix of on-premises/private and public cloud connected together.

Course 8: AWS Cloud - Practitioner to Solutions Architect Associate

Level	Beginner-Intermediate	**Duration**	8 weeks
Prerequisite	Cloud Fundamentals	**Modules**	12

Course Contents (All Modules)

- **Module 1:** AWS global infrastructure: regions, AZs, edge
- **Module 2:** IAM: users, roles, policies, MFA
- **Module 3:** EC2, AMIs, EBS, Auto Scaling, load balancers
- **Module 4:** S3, storage classes, lifecycle and versioning
- **Module 5:** VPC: subnets, route tables, IGW, NAT, security groups, NACLs
- **Module 6:** RDS, Aurora, DynamoDB
- **Module 7:** Route 53, CloudFront, API Gateway
- **Module 8:** Lambda, SQS, SNS, EventBridge

- **Module 9:** CloudWatch, CloudTrail, Config
- **Module 10:** CloudFormation and Well-Architected Framework
- **Module 11:** Cost management and billing
- **Module 12:** Exam preparation and practice labs

Tools and Platforms

AWS Console, CLI, CloudFormation, Terraform

Questions and Answers

Q1. What is the difference between a security group and a NACL?
A: Security groups are stateful and instance-level; NACLs are stateless and subnet-level.

Q2. Which S3 feature protects against accidental deletion?
A: Versioning (plus MFA Delete and Object Lock).

Q3. What is an Availability Zone?
A: One or more discrete data centres inside a region with independent power and networking.

Q4. When use Lambda?
A: For event-driven, short-running code without managing servers.

Course 9: Microsoft Azure Administrator (AZ-104 level)

Level	Intermediate	**Duration**	8 weeks
Prerequisite	Cloud Fundamentals	**Modules**	10

Course Contents (All Modules)

- **Module 1:** Azure architecture: subscriptions, resource groups, management groups
- **Module 2:** Microsoft Entra ID, RBAC and conditional access
- **Module 3:** Virtual machines, scale sets and availability options
- **Module 4:** Storage accounts, blobs, files, SAS tokens
- **Module 5:** Virtual networks, peering, VPN gateway, ExpressRoute
- **Module 6:** NSG, Azure Firewall, Bastion, Load Balancer, App Gateway
- **Module 7:** App Service, containers and Azure Functions
- **Module 8:** Backup, Site Recovery and monitoring
- **Module 9:** Azure Policy, Blueprints, cost management
- **Module 10:** ARM/Bicep templates and Azure CLI/PowerShell

Tools and Platforms

Azure Portal, Azure CLI, PowerShell, Bicep

Questions and Answers

Q1. What is a resource group?
A: A logical container that holds related Azure resources managed together.

Q2. What does RBAC do?
A: Grants permissions to users/groups at a given scope using roles.

Q3. What is an NSG?
A: A Network Security Group - rules filtering traffic to/from Azure resources.

Q4. Difference between Azure Backup and Site Recovery?
A: Backup protects data by restoring copies; Site Recovery replicates workloads for failover.

Course 10: Google Cloud Platform - Associate Cloud Engineer

Level	Intermediate	**Duration**	6 weeks
Prerequisite	Cloud Fundamentals	**Modules**	10

Course Contents (All Modules)

- **Module 1:** GCP resource hierarchy: org, folders, projects
- **Module 2:** IAM, service accounts and organisation policies
- **Module 3:** Compute Engine, instance groups, GKE
- **Module 4:** Cloud Storage, Filestore, Persistent Disk
- **Module 5:** VPC, firewall rules, Cloud NAT, VPN/Interconnect
- **Module 6:** Cloud SQL, Spanner, Bigtable, BigQuery
- **Module 7:** Cloud Run, App Engine, Cloud Functions
- **Module 8:** Pub/Sub and Dataflow
- **Module 9:** Operations suite: logging, monitoring
- **Module 10:** Terraform and gcloud CLI

Tools and Platforms

gcloud CLI, Cloud Shell, Terraform

Questions and Answers

Q1. What is a GCP project?
A: The base unit that owns resources, billing and APIs.

Q2. What is BigQuery?
A: A serverless, scalable data warehouse for SQL analytics.

Q3. What is a service account?
A: An identity used by applications/VMs instead of a human user.

Q4. Cloud Run vs GKE?
A: Cloud Run runs stateless containers serverlessly; GKE is managed Kubernetes for full orchestration control.

Course 11: Cloud Security (CCSP / AWS Security Specialty level)

Level	Advanced	**Duration**	8 weeks
Prerequisite	One major cloud + networking	**Modules**	11

Course Contents (All Modules)

- **Module 1:** Cloud security concepts and architecture
- **Module 2:** Identity and access management, federation, SSO
- **Module 3:** Data security: encryption at rest/in transit, KMS, HSM, DLP
- **Module 4:** Network security: segmentation, WAF, DDoS protection
- **Module 5:** Container and serverless security
- **Module 6:** Logging, SIEM and threat detection
- **Module 7:** Cloud misconfiguration and CSPM
- **Module 8:** Compliance: ISO 27001, SOC 2, GDPR, HIPAA, PCI DSS
- **Module 9:** Incident response and forensics in cloud
- **Module 10:** Zero Trust architecture

- **Module 11:** Secure DevOps (DevSecOps)

Tools and Platforms

Prowler, ScoutSuite, GuardDuty, Defender for Cloud, Security Command Center

Questions and Answers

Q1. What is the most common cause of cloud breaches?
A: Misconfiguration, such as public storage buckets and over-permissive IAM.

Q2. What is least privilege?
A: Granting only the permissions needed for a task.

Q3. What is CSPM?
A: Cloud Security Posture Management - continuous detection of misconfigurations and compliance gaps.

Q4. Why use customer-managed keys?
A: They give the customer control over key rotation, access and revocation.

Course 12: DevOps, Containers and Kubernetes

Level	Intermediate	**Duration**	8 weeks
Prerequisite	Linux + one cloud	**Modules**	10

Course Contents (All Modules)

- **Module 1:** DevOps culture and SDLC
- **Module 2:** Git, GitHub/GitLab workflows
- **Module 3:** CI/CD with Jenkins, GitHub Actions, GitLab CI
- **Module 4:** Docker: images, volumes, networking, registries
- **Module 5:** Kubernetes architecture: pods, services, deployments, ingress
- **Module 6:** Helm, ConfigMaps, Secrets, RBAC
- **Module 7:** Infrastructure as Code: Terraform and Ansible
- **Module 8:** Monitoring: Prometheus, Grafana, ELK
- **Module 9:** Container and pipeline security
- **Module 10:** Service mesh and GitOps (Argo CD)

Tools and Platforms

Docker, Kubernetes (minikube/kind), Terraform, Jenkins

Questions and Answers

Q1. What is the difference between a container and a VM?
A: Containers share the host kernel and are lightweight; VMs run a full guest OS on a hypervisor.

Q2. What is a Kubernetes pod?
A: The smallest deployable unit, containing one or more containers.

Q3. What is Infrastructure as Code?
A: Managing infrastructure through versioned, declarative code instead of manual steps.

Q4. What is GitOps?
A: Using Git as the single source of truth for deployments, applied automatically to clusters.

PART 3 - ETHICAL HACKING

Course 13: Ethical Hacking Foundations

Level	Beginner	**Duration**	4 weeks
Prerequisite	Networking Fundamentals	**Modules**	8

Course Contents (All Modules)

- **Module 1:** What is ethical hacking: white, grey and black hat
- **Module 2:** Laws, ethics, scope and written authorization
- **Module 3:** CIA triad, threats, vulnerabilities, risk
- **Module 4:** Attacker lifecycle: recon, scanning, access, persistence, covering tracks (to understand and defend)
- **Module 5:** Attack types overview: malware, phishing, DoS, MITM, injection
- **Module 6:** Security frameworks: MITRE ATT&CK, NIST, ISO 27001, CEH
- **Module 7:** Setting up a safe home lab (VirtualBox/VMware)
- **Module 8:** Career paths and certifications

Tools and Platforms

Kali Linux, VirtualBox, Metasploitable, DVWA

Questions and Answers

Q1. What separates an ethical hacker from a criminal hacker?
A: Written permission, defined scope, and the goal of improving security.

Q2. What is the CIA triad?
A: Confidentiality, Integrity and Availability.

Q3. Why is a scope document important?
A: It defines what may be tested, when, and how, protecting both tester and client legally.

Q4. What is MITRE ATT&CK;?
A: A knowledge base of real-world adversary tactics and techniques.

Course 14: Linux and Windows for Security Professionals

Level	Beginner	**Duration**	4 weeks
Prerequisite	None	**Modules**	9

Course Contents (All Modules)

- **Module 1:** Linux file system, permissions and users
- **Module 2:** Essential commands and shell scripting (Bash)
- **Module 3:** Package management and services
- **Module 4:** Windows architecture, registry, services
- **Module 5:** PowerShell basics and administration
- **Module 6:** User management, Group Policy, Active Directory basics
- **Module 7:** Logging: syslog, journald, Windows Event Logs
- **Module 8:** Hardening Linux and Windows systems
- **Module 9:** Python for security automation

Tools and Platforms

Kali, Ubuntu, Windows Server, PowerShell, Python

Questions and Answers

Q1. What does chmod 755 file mean?
A: Owner read/write/execute; group and others read/execute.

Q2. Where does Linux keep user account info?
A: /etc/passwd, with hashes in /etc/shadow.

Q3. What is Group Policy?
A: A Windows feature to centrally configure users and computers in a domain.

Q4. Why learn scripting?
A: To automate repetitive security tasks and analysis.

Course 15: Reconnaissance and Footprinting (OSINT)

Level	Beginner-Intermediate	**Duration**	3 weeks
Prerequisite	Ethical Hacking Foundations	**Modules**	9

Course Contents (All Modules)

- **Module 1:** Passive vs active reconnaissance
- **Module 2:** Search engine operators (Google dorking) for authorized assessments
- **Module 3:** WHOIS, DNS records, subdomain discovery
- **Module 4:** OSINT frameworks: Maltego, theHarvester, Recon-ng, SpiderFoot
- **Module 5:** Social media and public-record intelligence
- **Module 6:** Email and metadata analysis
- **Module 7:** Technology fingerprinting
- **Module 8:** Reducing your organization's digital footprint
- **Module 9:** Reporting recon findings

Tools and Platforms

theHarvester, Maltego, Shodan, Recon-ng, Amass

Questions and Answers

Q1. Passive vs active recon?
A: Passive gathers public info without touching the target; active interacts directly with target systems.

Q2. What can DNS records reveal?
A: Mail servers, subdomains, hosting providers and sometimes internal naming.

Q3. What is Shodan?
A: A search engine for internet-connected devices and exposed services.

Q4. How can organisations reduce footprint?
A: Remove stale subdomains, scrub metadata, restrict public info and train staff.

Course 16: Network Scanning and Enumeration

Level	Intermediate	**Duration**	4 weeks
Prerequisite	Networking + Linux	**Modules**	9

Course Contents (All Modules)

- **Module 1:** TCP/UDP, handshakes, ports and services
- **Module 2:** Host discovery techniques
- **Module 3:** Port scanning types with Nmap and masscan
- **Module 4:** Service and OS fingerprinting
- **Module 5:** Banner grabbing
- **Module 6:** Enumeration: SMB, SNMP, LDAP, DNS, NFS
- **Module 7:** Understanding IDS/IPS and firewall detection (defender view)
- **Module 8:** Packet analysis with Wireshark and tcpdump
- **Module 9:** Interpreting results and prioritising targets

Tools and Platforms

Nmap, Wireshark, Netcat, enum4linux

Questions and Answers

Q1. What does a SYN scan do?
A: Sends SYN packets and reads replies without completing the handshake.

Q2. Why is SNMP enumeration risky?
A: Default community strings can expose device and network details.

Q3. Which Nmap flag detects service versions?
A: -sV.

Q4. How do defenders detect scans?
A: IDS/IPS rules, log analysis and rate/anomaly alerts.

Course 17: Vulnerability Assessment and Management

Level	Intermediate	**Duration**	4 weeks
Prerequisite	Network Scanning	**Modules**	9

Course Contents (All Modules)

- **Module 1:** Vulnerability lifecycle and CVE/CVSS/CWE
- **Module 2:** Authenticated vs unauthenticated scanning
- **Module 3:** Nessus, OpenVAS/Greenbone, Qualys
- **Module 4:** Configuration auditing and CIS benchmarks
- **Module 5:** Prioritisation: CVSS, EPSS, business impact
- **Module 6:** Patch management and remediation workflows
- **Module 7:** False positive validation
- **Module 8:** Reporting for technical and executive audiences
- **Module 9:** Continuous vulnerability management programme

Tools and Platforms

OpenVAS, Nessus Essentials, Nmap NSE

Questions and Answers

Q1. What is CVSS?
A: A standard 0-10 scoring system for vulnerability severity.

Q2. Why run authenticated scans?
A: They see installed software and configs, giving more accurate results.

Q3. Vulnerability assessment vs penetration test?
A: An assessment identifies and ranks weaknesses; a pen test attempts controlled exploitation to prove impact.

Q4. What is a false positive?
A: A reported vulnerability that does not actually exist.

Course 18: Web Application Security and OWASP Top 10

Level	Intermediate	**Duration**	6 weeks
Prerequisite	HTTP basics + scripting	**Modules**	11

Course Contents (All Modules)

- **Module 1:** HTTP, cookies, sessions and web architecture
- **Module 2:** OWASP Top 10 overview
- **Module 3:** Injection (SQL, command, LDAP) and prevention
- **Module 4:** Broken authentication and session management
- **Module 5:** Cross-Site Scripting (XSS) and CSP
- **Module 6:** Broken access control, IDOR
- **Module 7:** CSRF, SSRF, security misconfiguration
- **Module 8:** Insecure deserialization and vulnerable components
- **Module 9:** API security (OWASP API Top 10)
- **Module 10:** Secure coding, input validation, WAF
- **Module 11:** Practice on legal labs: DVWA, Juice Shop, WebGoat

Tools and Platforms

Burp Suite, OWASP ZAP, DVWA, Juice Shop

Questions and Answers

Q1. How are SQL injection attacks prevented?
A: Parameterised queries/prepared statements, input validation and least-privilege database accounts.

Q2. What is XSS?
A: Injecting malicious scripts into pages viewed by other users; mitigated by output encoding and CSP.

Q3. What is IDOR?
A: Accessing other users' objects by changing an identifier because authorisation is missing.

Q4. What does a WAF do?
A: Filters and blocks malicious HTTP traffic before it reaches the app.

PART 4 - PENETRATION TESTING

Course 19: Penetration Testing Methodology and Planning

Level	Intermediate	**Duration**	3 weeks
Prerequisite	Ethical Hacking Foundations	**Modules**	9

Course Contents (All Modules)

- **Module 1:** Pen test types: black, grey, white box
- **Module 2:** PTES, OSSTMM, NIST SP 800-115, OWASP WSTG
- **Module 3:** Rules of engagement, scoping and contracts
- **Module 4:** Threat modelling and attack planning
- **Module 5:** Pre-engagement questionnaires and legal protections
- **Module 6:** Lab building: attack box, targets, VPN
- **Module 7:** Documentation and evidence handling
- **Module 8:** Reporting structure and risk rating
- **Module 9:** Retesting and remediation validation

Tools and Platforms

Kali Linux, Dradis, CherryTree, Obsidian

Questions and Answers

Q1. What is the PTES?
A: The Penetration Testing Execution Standard - a framework covering pre-engagement to reporting.

Q2. Black box vs white box?
A: Black box gives no prior knowledge; white box gives full details such as source and architecture.

Q3. What must be signed before testing?
A: A written authorisation/contract with scope and rules of engagement.

Q4. Why keep detailed notes?
A: They support reproducibility, accurate reports and legal accountability.

Course 20: Network Penetration Testing

Level	Intermediate-Advanced	**Duration**	8 weeks
Prerequisite	Scanning + Linux	**Modules**	10

Course Contents (All Modules)

- **Module 1:** External and internal test planning
- **Module 2:** Exploitation concepts using Metasploit in authorised labs
- **Module 3:** Password security: policies, hashing, cracking resistance
- **Module 4:** Network service weaknesses: SMB, FTP, SSH, RDP, SNMP
- **Module 5:** Man-in-the-middle and sniffing (lab/defence view)
- **Module 6:** Privilege escalation concepts on Linux and Windows
- **Module 7:** Pivoting and lateral movement concepts
- **Module 8:** Post-exploitation and persistence detection
- **Module 9:** Firewall and segmentation testing
- **Module 10:** Writing findings and remediation advice

Tools and Platforms

Metasploit, Nmap, John, Hashcat (lab use only)

Questions and Answers

Q1. Why do weak passwords remain a top finding?
A: Users reuse or choose guessable passwords, enabling guessing and cracking.

Q2. What is privilege escalation?
A: Gaining higher permissions than originally obtained.

Q3. What is pivoting?
A: Using a compromised host to reach otherwise inaccessible network segments.

Q4. Best defence for lateral movement?
A: Segmentation, least privilege, MFA, patching and monitoring.

Course 21: Web Application Penetration Testing

Level	Advanced	**Duration**	8 weeks
Prerequisite	Web App Security	**Modules**	11

Course Contents (All Modules)

- **Module 1:** Testing methodology: OWASP WSTG
- **Module 2:** Mapping the application and attack surface
- **Module 3:** Authentication and session testing
- **Module 4:** Authorisation testing and business logic flaws
- **Module 5:** Input validation: injection, XSS, file upload, XXE
- **Module 6:** API and GraphQL testing
- **Module 7:** Client-side testing: DOM, CORS, JWT
- **Module 8:** Using Burp Suite extensions and automation
- **Module 9:** Chaining vulnerabilities for impact
- **Module 10:** Secure remediation guidance
- **Module 11:** Bug bounty basics and responsible disclosure

Tools and Platforms

Burp Suite, ZAP, ffuf, Postman

Questions and Answers

Q1. What is responsible disclosure?
A: Privately reporting a vulnerability to the owner and allowing time to fix before publishing.

Q2. What is a business logic flaw?
A: A design flaw that lets users misuse legitimate functionality, e.g. skipping payment steps.

Q3. Why is JWT testing important?
A: Weak signing, algorithm confusion or no expiry can enable token forgery.

Q4. What is CORS misconfiguration?
A: Overly permissive cross-origin rules letting malicious sites read sensitive responses.

Course 22: Wireless and Mobile Penetration Testing

Level	Advanced	**Duration**	6 weeks
Prerequisite	Wireless Networking + Network Pen Testing	**Modules**	10

Course Contents (All Modules)

- **Module 1:** Wireless threat landscape and legal considerations
- **Module 2:** Wi-Fi encryption weaknesses: WEP, WPA/WPA2 (own lab)
- **Module 3:** Rogue AP and evil twin detection
- **Module 4:** Enterprise Wi-Fi 802.1X weaknesses
- **Module 5:** Bluetooth and IoT radio basics
- **Module 6:** Android app architecture, APK analysis
- **Module 7:** iOS app security model
- **Module 8:** OWASP MASVS and Mobile Top 10
- **Module 9:** Insecure storage, communications and authentication in apps
- **Module 10:** Reporting and hardening guidance

Tools and Platforms

Kismet, MobSF, Frida, Burp (own lab)

Questions and Answers

Q1. Why is WPA3 stronger?
A: It uses SAE, protecting against offline dictionary attacks and giving forward secrecy.

Q2. What is an evil twin?
A: A rogue access point imitating a legitimate network to lure users.

Q3. What does MobSF do?
A: Automates static and dynamic analysis of mobile apps.

Q4. Biggest mobile app flaw?
A: Insecure data storage and weak server-side authorisation.

Course 23: Cloud Penetration Testing

Level	Advanced	**Duration**	6 weeks
Prerequisite	Cloud Security + Pen Testing basics	**Modules**	10

Course Contents (All Modules)

- **Module 1:** Cloud pen testing rules and provider policies
- **Module 2:** Cloud attack surface and recon
- **Module 3:** IAM misconfiguration and privilege escalation paths
- **Module 4:** Storage exposure: buckets, blobs, snapshots
- **Module 5:** Metadata services and SSRF risks
- **Module 6:** Serverless and container testing
- **Module 7:** Kubernetes misconfiguration testing
- **Module 8:** Azure Entra ID and AWS IAM attack paths
- **Module 9:** Detection engineering and logging validation
- **Module 10:** Reporting and remediation

Tools and Platforms

Pacu, ScoutSuite, Prowler, CloudFox, kube-hunter, Trivy

Questions and Answers

Q1. Do you need provider permission to test cloud?
A: Follow the provider's pen test policy; many allow common services without prior approval, but rules vary.

Q2. Why is the metadata service a target?
A: SSRF can expose instance credentials; IMDSv2 mitigates this.

Q3. Common cloud privilege escalation cause?
A: Over-permissive IAM policies such as wildcard actions or PassRole misuse.

Q4. What does Trivy scan?
A: Container images, file systems and IaC for known vulnerabilities and misconfigurations.

Course 24: Active Directory Security, Red Team Concepts and Reporting

Level	Advanced	**Duration**	8 weeks
Prerequisite	Network Pen Testing	**Modules**	10

Course Contents (All Modules)

- **Module 1:** Active Directory architecture: forests, domains, GPO
- **Module 2:** AD enumeration with BloodHound (authorised lab)
- **Module 3:** Kerberos, NTLM and common weaknesses
- **Module 4:** Credential hygiene, tiering and LAPS
- **Module 5:** Red team vs pen test vs purple team
- **Module 6:** MITRE ATT&CK mapping and adversary emulation
- **Module 7:** Understanding EDR evasion (defender perspective)
- **Module 8:** Blue team detection: SIEM, Sysmon, Sigma rules
- **Module 9:** Professional report writing and presentation
- **Module 10:** Careers: OSCP, PNPT, CEH, GPEN, CRTP

Tools and Platforms

BloodHound, Sysmon, Splunk/ELK, Atomic Red Team, Caldera

Questions and Answers

Q1. What is Kerberoasting at a high level?
A: Requesting service tickets for accounts and cracking weak service-account passwords offline.

Q2. How to defend AD?
A: Tiered admin model, strong service-account passwords/gMSA, MFA, patching and monitoring.

Q3. What is purple teaming?
A: Red and blue teams collaborating to improve detection and response.

Q4. What goes in a good pen test report?
A: Executive summary, scope, methodology, findings with risk, evidence, remediation and retest notes.

Recommended Learning Paths

Goal	Suggested order (course numbers)
Network Engineer	1 > 2 > 3 > 5 > 4 > 6
Cloud Engineer / Architect	1 > 7 > 8 or 9 or 10 > 12 > 11
Ethical Hacker (entry)	1 > 13 > 14 > 15 > 16 > 17 > 18
Penetration Tester	1 > 2 > 13 > 14 > 16 > 18 > 19 > 20 > 21
Cloud Security Specialist	7 > 8 > 11 > 23
Red Team / Advanced	19 > 20 > 21 > 22 > 23 > 24

Certification Roadmap

Area	Beginner	Intermediate	Advanced
Networking	CompTIA Network+	Cisco CCNA	Cisco CCNP / CCIE
Cloud	AWS Cloud Practitioner, AZ-900	AWS SAA, AZ-104, GCP ACE	AWS Pro, CCSP, CKA/CKS
Ethical Hacking	CompTIA Security+	CEH, eJPT	CISSP, CASP+
Pen Testing	eJPT, PenTest+	PNPT, GPEN	OSCP, OSEP, CRTP

General FAQ

Q1. Do I need programming to start?
A: No. Start with networking and Linux; add Python and Bash as you progress.

Q2. What hardware do I need for a lab?
A: A PC with 16 GB RAM, a virtualisation tool (VirtualBox/VMware) and free cloud accounts is enough.

Q3. How long to become job-ready?
A: Typically 6 to 12 months of consistent study and labs for entry-level roles.

Q4. Is a degree required?
A: Not strictly; certifications, labs, CTF results and a portfolio carry strong weight.

Q5. Where can I practise legally?
A: TryHackMe, Hack The Box, PortSwigger Web Security Academy, OverTheWire, cloud free tiers and your own home lab.

Q6. Which should I learn first: cloud or security?
A: Networking and Linux first; then cloud or security depending on your career goal.

Q7. What is a CTF?
A: Capture The Flag - a legal competition where you solve security challenges to find hidden flags.

Q8. What is the difference between ethical hacking and penetration testing?
A: Ethical hacking is the broad discipline; penetration testing is a structured, scoped engagement with a formal report.